Making Flock Zero-Knowledge:

Fixed-digest BLAKE3, symbolic coverage, and an executable ROM game

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Abstract

Flock's original BLAKE3 prover reveals deterministic functions of its witness. This note specifies and implements a zero-knowledge mode for two exact production profiles: the free BLAKE3 batch relation and, more importantly, the fixed-digest relation

$$y_i = \mathrm{BLAKE3}(x_i), \qquad |x_i| = 64 \text{ bytes}, \quad 1 \le i \le 256.$$

The construction uses field-valued masking over $\mathbb{F}_{2^{128}}$, a public affine $Q^\star$, a hiding Ligerito commitment, a proof nonce, and one framed random oracle shared by Fiat–Shamir, Merkle trees, and proof of work. A symbolic coverage calculation emits a nonzero minor of total-degree bound 720; hence the production PIOP bad set has probability at most $720/2^{128}$. A closed-form PCS translator preserves the combined opening vector and leaves 16,384 bits of conditional entropy in each fresh initial wide leaf. A sealed, one-pass simulator sees only the public digest statement, programs at most 18 fresh oracle points, and produces a proof accepted by the unmodified verifier.

Under the classical programmable-random-oracle model and the assumptions made explicit below, this gives *computational zero knowledge* for the two registered circuit digests. At an adversarial hash-query budget $Q = 2^{64}$, the symbolic PIOP term has 118.508 bits and the explicit listed union bound is 118.502 bits. Knowledge is a separate claim: the shipped standalone profile has about 55.994 bits of currently provable classical knowledge security after the Fiat–Shamir loss and is labelled *100-bit conjectured classical knowledge security.* It does not meet a 100-bit standalone theorem threshold.

1 Scope and security labels

The certificate registry contains exactly the following profiles. Unknown families, circuit digests, batch sizes, or PCS parameters are rejected before proof generation.

Family	Public input	Circuit / PCS	ZK status
BLAKE3 batch	batch size 256	$m = 22$, rate 1/2, batch log 6	computational ZK
BLAKE3 preimage	256 digests	$m = 22$, rate 1/2, batch log 6	computational ZK

The preimage family covers one full BLAKE3 hash of exactly 64 bytes. Its R1CS pins the IV, counter, block length, and `CHUNK_START`, `CHUNK_END`, and `ROOT` flags. SHA-256, Keccak, longer BLAKE3 messages, chains, Merkle-path statements, recursion, side channels, and the QROM are out of scope. No claim in this note transfers to those cases.

Mandatory separation of claims. "Computational ZK" describes witness hiding. "Knowledge security" describes extraction. The shipped standalone profile may be described as *100-bit conjectured classical knowledge security*; its theorem column must remain 55.994 bits at $Q = 2^{64}$. Section 8 gives the three permitted knowledge profiles.

Assumption 1 (Classical programmable random oracle)**.** SHA-256, used with the injective frames in Section 3, is modelled as a classical random oracle. The simulator may program fresh, previously unqueried points. Collision resistance and pseudorandomness hide unopened Merkle leaves and siblings. Quantum queries are excluded.

2 The relation and protocol

Let z be the packed binary assignment for the block-diagonal BLAKE3 R1CS, embedded into $\mathbb{F}_{2^{128}}$. The prover establishes

$$(Az) \circ (Bz) = Cz$$

with Flock's zerocheck and lincheck, then opens the committed vector using the ring-switched Ligerito PCS. In the fixed-digest family the verifier also derives a packed-direct opening claim from the public digest list. The list, padding policy, witness layout, and circuit matrices are transcript-bound before the first challenge.

The zero-knowledge prover adds independent randomizer rows and four committed mask sources: the field-valued zerocheck mask P, the lincheck mask S, and the round-one masks S_c, S_h. The witness commitment and each mask commitment use independent μ and g hiding randomness. All streams, including the proof nonce, are forks of one per-proof DRBG under distinct labels.

The zerocheck mask is

$$\gamma P(X) Q^\star(X), \qquad Q^\star(X) = q_0 + \sum_j q_j X_j,$$

where P is uniform over $\mathbb{F}_{2^{128}}$ on a translated subcube of the top $d_{\log} = 12$ variables and $Q^\star$ is protocol-fixed, public, affine, and nonzero on every Boolean cube point. This replaces the withdrawn construction with two Boolean masks P, Q. There is no Q commitment, opening, final evaluation, or mask-randomness fork in protocol `flock-zk-fv-v3`.

3 One framed random oracle

Every hash point begins with a disjoint role byte. Merkle points additionally bind protocol version, proof nonce, commitment channel, recursion depth, node level, node index, payload length, and payload. Proof-of-work points are

$$\texttt{ROLE\_POW} \,\|\, \textit{state} \,\|\, \textit{nonce}.$$

Fiat–Shamir absorbs the 32-byte proof nonce immediately after the statement and before any challenge. Native hashing and external recording/programming backends consume identical frames and return byte-identical answers when no point is programmed.

These frames serve three purposes. First, the Merkle extractor can recognize every witness leaf query without relying on observation order. Second, simulator programming cannot alias a leaf, internal node, transcript squeeze, or proof-of-work query. Third, reusing a proof across nonces, commitment channels, or recursive depths changes every relevant oracle point. Scalar and four-way SIMD midstate implementations are tested against the one-shot definition for every tail shape.

4 Symbolic PIOP coverage

Fix the challenges after the commitment. Let R map the coefficients of P to the $2n$ zerocheck round-message coordinates, and let L map them to the two mask values revealed later: the initial masked claim and $P(\rho)$. For a fixed leakage value the available displacement is $R(\ker L)$.

Theorem 1 (Conditioned field-mask coverage)**.** *For $m \in \{13, 15, 22\}$, the emitted symbolic matrix satisfies, outside its selected minor's zero set,*

$$\operatorname{rank} \begin{bmatrix} R \\ L \end{bmatrix} = 2n, \qquad \operatorname{rank} L = 2,$$
$$\dim R(\ker L) = 2n - 2.$$

At the production shape $m = 22$, the selected determinant has total degree at most 720 in the 32 challenge variables. Uniform challenges therefore give

$$\Pr[\textit{coverage failure}] \le \frac{720}{2^{128}} < 2^{-118.508}.$$

The implementation derives both the concrete and symbolic matrices from the same generic round kernels. `SymScalar` has three backends: concrete $\mathbb{F}_{2^{128}}$ evaluation, conservative per-variable tropical degree bounds, and exact sparse multivariate polynomials for toy instances. It deliberately exposes no inversion operation. Thus challenge-dependent denominators cannot silently enter the polynomial calculation.

The S2 JSON artifact pins the selected rows, columns, nonzero determinant evaluations, and degree vectors. One Lean adapter instantiates Mathlib's sum-of-per-variable-degree Schwartz–Zippel theorem; another proves the bad-set mixture step. This theorem covers the field-mask block; it does not by itself claim PCS opening privacy.

5 PCS replacement and entropy

Write the initial hiding commitment's combined vector as

$$F = (\mu \parallel z) + c\, g,$$

where $c \in \mathbb{F}_{2^{128}}$ is transcript-derived. For any claim-preserving witness difference d and $c \neq 0$, the translator sets

$$\delta g_{\mathrm{top}} = c^{-1} d$$

so that $\delta F_{\mathrm{top}} = 0$. It solves the selected initial-level encoding equations for $\delta\mu$ and takes $\delta g_{\mathrm{lo}} = c^{-1}\delta\mu$, making every opened initial wide row identical as well. Characteristic two removes sign distinctions in these equalities.

The production opening makes 218 initial-level queries while each lane has 512 independent mask symbols. After conditioning on an opened wide leaf, at least 16,384 bits remain. Since $\delta F = 0$ globally, every recursive codeword, root, sibling, sumcheck value, and out-of-domain sample is identical; no recursive entropy charge is needed. The only algebraic exceptional event is $c = 0$, with probability 2^{-128}.

The implementation exhaustively destructures all opening proof fields into a 12-category manifest. Adding a proof field without classifying it is a compile failure. The exact translator is differentially tested against the shipped NTT and wide interleaving. The pinned artifacts are `s3_opening_functionals.json` and `s3_minentropy_table.json`.

6 The sealed simulator game

The simulator accepts a capability `SealedStatement` whose private fields contain only the setup and public digests; its function type cannot name a witness or preimage. In one pass it constructs a trace for unrelated messages, overwrites the public-output region, samples the zerocheck terminal values, and solves the dependent messages. The patched trace is not a valid R1CS witness, and the honest prover run on it is rejected. Nevertheless, after programming the required fresh oracle points, the unmodified production verifier accepts the simulated proof.

The executable game ledger consists of:

Game	Change	Charged term
G_0	honest proof	0
G_1	translate PIOP masks	$720/2^{128}$
G_2	translate μ, g	$\Pr[c = 0] + \varepsilon_{\text{solve}}$
G_3	replace hidden Merkle leaves	$\varepsilon_{\text{sibling}} + \varepsilon_{\text{hash}}$
G_4	pre-sample and program challenges	$\varepsilon_{\text{prequery}} + \varepsilon_{\text{prefix}}$
G_5	sealed one-pass simulator	degenerate-challenge terms

At $Q = 2^{64}$ adversarial oracle queries and at most 18 programmed points, the prequery term has 187.830 bits, the initial sibling term has more than 16,300 bits, and the symbolic rank term has 118.508 bits. Summing the listed terms gives 118.502 bits before conventional hash-collision terms.

Theorem 2 (Scoped computational zero knowledge)**.** *Assume the classical programmable-RO model, Theorem 1, the PCS translator and entropy bounds of Section 5, and freshness of the programmed points. Then the registered fixed-digest profile is computationally zero knowledge, with distinguishing advantage bounded by the sum of the six game terms above. The registered free-batch profile follows as the easier self-generated-witness special case.*

This is a classical-ROM theorem. It is neither a QROM result nor a claim that SHA-256 is an information-theoretic random oracle.

7 Extraction and simulation extractability

The point-oracle recorder observes all framed leaf queries made while building the witness commitment. For a complete valid codeword it reconstructs the interleaved word, applies the inverse NTT lane by lane, re-encodes the candidate, checks the configured unique-radius inequality, and returns the witness half; the low mask and blinder lanes are separate outputs. The system-specific extractor then reads exactly the 512 message bits of each instance and checks

$$\text{BLAKE3}(x_i) = y_i$$

with the external `blake3` implementation. This final check refers to no circuit wire or proof claim. Honest recorded commitments recover the real preimages; the simulator's commitment fails extraction.

The executable decoder is a straight-line candidate validator and exact-word decoder, not a complete Reed–Solomon error-correction implementation for every received word within the theoretical radius. The general noisy-word step is therefore supplied by the Ligerito proximity/unique-decoding theorem, not by a claim that the current routine corrects arbitrary errors.

For simulation extractability, the simulator records the prefix

$$(\textit{statement digest}, \textit{proof nonce}, \textit{protocol version}).$$

Extraction is allowed only on a fresh prefix and refuses a prefix returned by the simulation oracle. This is *fresh-prefix weak simulation extractability.* Same-prefix simulation extractability and straight-line random-beacon-resistant knowledge are not claimed.

8 Knowledge security profiles

The interactive opening has an L0 fold access-structure term at least $257/2^{128}$, corresponding to 119.994 bits. The classical Fiat–Shamir knowledge reduction charges $(Q+1)\kappa_\Gamma$. At $Q = 2^{64}$, this term alone limits the standalone theorem to

$$128 - \log_2 257 - 64 = 55.994 \text{ bits.}$$

This loss is classical, not post-quantum.

Profile	Provable bits	Required label
Standalone FS over $\mathbb{F}_{2^{128}}$	55.994 at $Q = 2^{64}$	100-bit conjectured classical knowledge security
Unbiased post-commitment beacon over $\mathbb{F}_{2^{128}}$	119.994	provable 100-bit; not standalone
Standalone FS with challenge-only $\mathbb{F}_{2^{256}}$	183.994 at $Q = 2^{64}$	engineering follow-up; machinery not implemented

The repository pins both the interactive and post-Fiat–Shamir columns in `docs/artifacts/knowledge-ledger.json`. Running `scripts/knowledge-ledger.py` with `--check` regenerates and compares them. Grinding, a small-Q convention, generic parallel repetition, or omitting an individual fold challenge does not repair the standalone bound.

9 Implementation, reproduction, and limits

The wire protocol is `flock-zk-fv-v3`; proof I/O version 6 and transcript schema version 5 bind the proof nonce, framed oracle domains, and P-only mask schema. The exact registered circuit digests are:

```
BLAKE3 batch:
4a398ca2e73d9d1f70611bd6a2a408b0404aabddcf6cf75c29cfa412f72f8423
BLAKE3 fixed digest:
e71bde2b053b1b697412d2d30c2e2e57f7b5f1822484fcbcc15c89ea202890d5
```

The certification command is fail-fatal and rejects zero-test filters:

```
rustup override set stable
scripts/zk-certify.sh
cargo test --release --workspace --features zk,symbolic
cargo clippy --workspace --all-targets --features zk,symbolic -- -D warnings
cd lean && lake build
```

It replays the symbolic artifacts, transcript schema, RO framing, production rank gates, simulator, extractor, fixed-digest and batch roundtrips, Lean adapters, knowledge ledger, and the ban on direct SHA-256 calls outside the two reviewed oracle modules. Certificate evidence names are compared with script test names in both directions.

The following limitations remain material:

- The proof relies on the classical programmable-RO abstraction and has no QROM analysis.
- The knowledge reduction does not prove 100 standalone bits over $\mathbb{F}_{2^{128}}$; the precise label in Section 8 is mandatory.
- The executable decoder validates a candidate but does not implement a complete arbitrary-error RS decoder.
- Only 64-byte BLAKE3 preimages and the two exact registry entries are enabled. Other Flock statement families remain non-zk.
- Side-channel resistance, recursive composition, and same-prefix simulation extractability are out of scope.
- The construction and proof have not received independent cryptographic review. The executable evidence is a reproducibility aid, not a substitute for review.

Subject to these limits, the implementation now has the object absent from the earlier drafts: a public-statement-only simulator whose output is accepted by the unmodified verifier, together with a symbolic distribution argument and a fail-closed registry for the exact deployed shapes. That is the scope in which Flock is zero knowledge.

References

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